MICROSOFT AZURE FUNDAMENTALS (AZ-900)

Advanced Practice Questions & Answer Key

Comprehensive 45-Question Architectural, Governance, Security, and Pricing Study Reference

1. An enterprise requires absolute data residency isolation where compute resources must run on physical hardware dedicated solely to a single customer, and virtual machine instances must not share hardware with any other tenant. Which deployment model or feature satisfies this architectural constraint?

> **Correct Answer:** Azure Dedicated Host

2. A multinational corporation needs to deploy a cloud resource hierarchy in Azure. Which statement accurately describes the inheritance behavior of Azure policy definitions and Role-Based Access Control (RBAC) assignments?

> **Correct Answer:** Policies and RBAC assignments applied at a management group level are automatically inherited by all child management groups and subscriptions.

3. When configuring network security rules in Azure, what is the default evaluation priority and processing behavior for Network Security Groups (NSGs)?

> **Correct Answer:** Rules are processed in order of priority numbers from lowest to highest, stopping processing immediately when a matching rule condition is met.

4. Which Azure storage tier is optimized for data that is accessed infrequently and requires a minimum storage duration of at least 30 days?

> **Correct Answer:** Cool storage tier

5. An enterprise needs to ensure that critical Azure virtual machines remain highly available during scheduled physical host updates or unplanned hardware failures within a region. Which Azure feature protects VMs against multi-tenant infrastructure failures?

> **Correct Answer:** Availability Sets with fault domains and update domains

6. What is the primary architectural purpose of deploying Azure Resource Manager (ARM) templates or Bicep files?

> **Correct Answer:** To enable declarative infrastructure-as-code deployments for repeatable and consistent resource provisioning.

7. When designing disaster recovery across Azure regions, what is the fundamental architectural differentiator between Geographically Redundant Storage (GRS) and Read-Access Geographically Redundant Storage (RA-GRS)?

> **Correct Answer:** RA-GRS allows read access to data in the secondary region even if the primary region is completely unavailable, whereas standard GRS does not.

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8. Which identity management feature in Microsoft Entra ID enables organizations to require multi-factor authentication based on real-time signals such as user risk, device state, and location?

> **Correct Answer:** Microsoft Entra Conditional Access

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9. What is the primary architectural function of an Azure Load Balancer operating at layer 4 of the OSI model?

> **Correct Answer:** It distributes incoming TCP and UDP traffic flows across backend pool instances based on source and destination IP addresses and ports.

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10. Which Azure management service provides a unified security management system and advanced threat protection across hybrid cloud workloads and datacenters?

> **Correct Answer:** Microsoft Defender for Cloud

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11. An organization wants to connect its on-premises datacenter to an Azure Virtual Network with maximum reliability, high throughput, and encryption across the public internet. Which connectivity model should they choose?

> **Correct Answer:** Azure VPN Gateway with an IPsec/IKE site-to-site tunnel

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12. What is the primary benefit of deploying Azure Virtual Network (VNet) Peering?

> **Correct Answer:** It connects virtual networks seamlessly with low latency and high bandwidth through Microsoft's backbone network without traversing the public internet.

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13. Which Azure support plan level includes a dedicated Technical Account Manager (TAM) and 24/7 access to support engineers with a 15-minute response time for critical business-impact incidents?

> **Correct Answer:** Premier / Enterprise support plan

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14. An architect needs to ensure that no developer can create virtual machines with public IP addresses within a specific Azure subscription, regardless of their RBAC permissions. Which Azure service enforces this governance guardrail?

> **Correct Answer:** Azure Policy

15. What is the primary operational distinction between Azure Functions and Azure App Service Web Apps?

> **Correct Answer:** Azure Functions is an event-driven serverless compute service executing short-lived code snippets, whereas Web Apps host continuous web applications.

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16. Which Azure storage redundancy option replicates data synchronously across three availability zones within a single primary region?

> **Correct Answer:** Zone-redundant storage (ZRS)

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17. What is the primary role of an Azure Management Group?

> **Correct Answer:** To provide a scope level above subscriptions for efficient management of access, policies, and compliance across multiple subscriptions.

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18. When utilizing Azure Cosmos DB, which consistency model provides the absolute strongest consistency guarantees at the expense of higher write latency and lower availability during network partitions?

> **Correct Answer:** Strong consistency

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19. What is the primary function of Azure Bastion in an enterprise cloud architecture?

> **Correct Answer:** It provides secure, fully managed RDP and SSH connectivity to virtual machines directly over SSL (TLS) without exposing public IP addresses on the VMs.

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20. Which Azure cost management tool allows users to forecast future cloud expenditures, set spending thresholds, and receive automated alerts when budgets are exceeded?

> **Correct Answer:** Azure Budgets and Cost Management

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21. What is the architectural role of Azure Front Door in global application delivery?

> **Correct Answer:** It is a modern global cloud content delivery network (CDN) and Layer 7 load balancer providing fast, reliable, and secure access to web applications.

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22. An administrator needs to store sensitive application connection strings and API keys securely, ensuring they are encrypted at rest with keys managed by the organization in an HSM-backed vault. Which Azure service should be used?

> **Correct Answer:** Azure Key Vault

23. Which Azure compute service allows developers to run containerized applications without managing underlying virtual machines or Kubernetes clusters, scaling containers instantly based on HTTP traffic or event queues?

> **Correct Answer:** Azure Container Apps

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24. What is the primary function of Azure Arc in hybrid and multi-cloud architectural strategies?

> **Correct Answer:** It extends Azure management and governance to resources running on-premises, edge locations, or other clouds.

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25. When configuring network isolation, what is the key difference between a Network Security Group (NSG) and Azure Firewall?

> **Correct Answer:** An NSG filters subnet/NIC traffic at layer 3/4, whereas Azure Firewall is a stateful, fully managed cloud firewall supporting FQDN-based application rules and threat intelligence.

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26. What is the primary purpose of Microsoft Entra Privileged Identity Management (PIM)?

> **Correct Answer:** To enable time-bound, approval-based activation of administrative roles to mitigate risks of standing privileged access.

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27. Which Azure diagnostic and monitoring service ingests log data and telemetry from any source across Azure and on-premises, providing a powerful Kusto Query Language (KQL) query engine for analysis?

> **Correct Answer:** Azure Monitor Log Analytics workspace

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28. An enterprise requires that all Azure resources created within a specific region must be tagged with a 'Department' metadata label. Which Azure feature enforces this standard?

> **Correct Answer:** Azure Policy with 'Modify' or 'Append' or 'Deny' effects

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29. What is the primary architectural advantage of using Azure Availability Zones?

> **Correct Answer:** They protect applications and data from datacenter-level failures by providing physically separate datacenters with independent power, cooling, and networking within a region.

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30. Which Azure storage redundancy option provides the lowest cost by storing data locally within a single physical datacenter without replication to other datacenters or regions?

> **Correct Answer:** Locally redundant storage (LRS)

31. What is the primary operational characteristic of Azure Logic Apps?

> **Correct Answer:** A cloud service that helps automate and orchestrate tasks, business processes, and workflows when you need to integrate apps, data, systems, and services.

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32. When planning an Azure migration, which tool assesses on-premises VMware, Hyper-V, and physical server environments to evaluate migration suitability, sizing, and cost estimates?

> **Correct Answer:** Azure Migrate

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33. What is the primary function of Azure Service Health?

> **Correct Answer:** It provides a personalized view of the health of Azure services, regional outages, and planned maintenance affecting your specific resources.

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34. Which Azure database service offers fully managed SQL Server instances in the cloud with 100% compatibility with on-premises SQL Server Enterprise Edition?

> **Correct Answer:** Azure SQL Managed Instance

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35. What is the primary security benefit of implementing Microsoft Entra Multi-Factor Authentication (MFA)?

> **Correct Answer:** It mitigates the risk of compromised credentials by requiring two or more verification methods for user authentication.

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36. When configuring a storage account, what is the purpose of setting a soft delete policy for blob storage?

> **Correct Answer:** It enables recovery of deleted blobs or snapshots within a specified retention period before permanent deletion.

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37. What is the primary architectural function of an Azure Application Gateway?

> **Correct Answer:** It is a Layer 7 web traffic load balancer that supports HTTP/HTTPS traffic, SSL termination, and cookie-based session affinity.

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38. Which Azure feature allows organizations to test new Azure capabilities and preview upcoming features before they are released generally to the public?

> **Correct Answer:** Azure Preview features

39. An administrator needs to deploy a scalable web application that automatically provisions and scales virtual machines based on CPU utilization metrics or a defined time schedule. Which Azure service meets this requirement?

> **Correct Answer:** Azure Virtual Machine Scale Sets

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40. What is the primary role of the Microsoft Trust Center?

> **Correct Answer:** It is a website providing information and resources regarding security, privacy, compliance, and transparency across Microsoft cloud products.

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41. Which Azure database service is designed as a fully managed relational database service built on open-source MySQL and PostgreSQL engines?

> **Correct Answer:** Azure Database for MySQL and PostgreSQL

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42. What is the primary benefit of utilizing Azure Content Delivery Network (CDN) for web applications?

> **Correct Answer:** It caches static web assets on strategically placed edge nodes globally to reduce latency and accelerate content delivery to users.

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43. Which Azure support plan is available free of charge to all Azure subscribers and includes billing and subscription management support?

> **Correct Answer:** Basic support plan

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44. What is the primary architectural function of Azure Traffic Manager?

> **Correct Answer:** It is a DNS-based traffic load balancer that distributes incoming user requests across global Azure regions using routing methods such as priority or performance.

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45. When deploying resources in Azure, what is the hierarchical relationship between Management Groups, Subscriptions, Resource Groups, and Resources?

> **Correct Answer:** Management Groups can contain multiple Subscriptions, Subscriptions contain multiple Resource Groups, and Resource Groups contain multiple Resources.